Critical dependency unavailable

Northwind Logistics (fictional) - Scenario SCN-D-DEPENDENCY-UNAVAILABLE - Incident 2026-10-20 09:00 UTC - Data as of 2026-10-20 09:00 UTC

Data source: Bundled synthetic baseline - Analysis 2026.10.1 - Generated 2026-10-28 07:09 UTC

1. Executive summary

- Recovery readiness is 68.6/100 (Weak). 7 of 13 systems have evidence of a recent successful end-to-end restore test.
- The recovery plan covers 2 system(s). Estimated time to restore: Cannot be estimated: every recovery step is blocked.
- Potential data-loss exposure for systems being restored: 7 h; no usable recovery point for db-orders (loss not bounded).
- 2 critical and 7 high-severity findings were raised. 4 system(s) would lose more data than their RPO allows; 2 plan step(s) miss their RTO.
- Blocked: app-orders-api, db-orders. These cannot be restored with the current evidence and dependencies.
- Systems affected by the incident (directly or through dependencies): app-erp, app-orders-api, db-orders, web-storefront.

Metric	Value	Note
Recovery readiness	68.6 / 100	Weak
Restore-tested systems	7 / 13	evidence-backed end-to-end tests
Stale or missing recovery points	4	RPO breached or no eligible point
Integrity failures	1	recovery points failing verification
Unmet RTO	2	0 unknown
Unmet RPO	4	1 unknown
Dependency failures	1	blocked by prerequisites, cycles or gaps
Business capability impacted	37.5%	criticality-weighted

2. Incident description

At 08:30 UTC ransomware encrypts the Orders API servers. The API's own backups are intact. While investigating, responders find the attacker had disabled and scheduled deletion of the customer-managed key that encrypts the orders database backups (08:35 UTC). The database is down and its recovery points cannot be decrypted. Restoring the API alone will not bring the service back.

Time (UTC)	Injected condition (simulated)	Detail
2026-10-20 08:30 UTC	1 asset(s) compromised	Ransomware encrypted the Orders API application servers.
2026-10-20 08:35 UTC	Asset(s) unrecoverable	Backup encryption key disabled and pending deletion; recovery points cannot be decrypted. Attacker disabled the customer-managed key for database backups. The database instance was also terminated.

3. Assessment scope and asset impact

13 protected systems assessed. Plan scope: systems taken down by the incident. Business capability impacted: 37.5% (criticality-weighted).

System	Tier	State	Being restored	Blocked
Orders Database (PostgreSQL)	tier0	unavailable	yes	cannot be restored: Backup encryption key disabled and pending deletion; recovery points cannot be decrypted
Orders API Service	tier0	compromised	yes	waiting on blocked prerequisite(s): db-orders
Customer Storefront	tier0	operational	no (dependency)	no
Internal ERP	tier1	operational	no (dependency)	no

4. Recovery readiness, backup findings and RTO/RPO compliance

System	Tier	Evidence	Selected point	Data loss	RPO	Ready	RTO	Score
Core DNS & Network Services	tier0	Restore-tested	RP-DNS-202610200200-P	7 h	met (1 d)	n/a	not applicable (30 min)	100
Identity Provider & SSO	tier0	Restore-tested	RP-IDP-202610200830-P	30 min	met (1 h)	n/a	not applicable (1 h)	100

System	Tier	Evidence	Selected point	Data loss	RPO	Ready	RTO	Score
Secrets & Key Management	tier0	Restore-tested	RP-KMS-202610200800-P	1 h	met (4 h)	n/a	not applicable (45 min)	100
Orders Database (PostgreSQL)	tier0	No usable point	none eligible	not bounded	breached (15 min)	blocked	breached (2 h)	25
Customer Database (PostgreSQL)	tier0	Integrity only	RP-DBCUS-202610200800-P	1 h	met (1 h)	n/a	not applicable (2 h)	66
Orders API Service	tier0	Restore-tested	RP-ORDAPI-202610200200-P	7 h	met (1 d)	blocked	breached (2 h 30 min)	65
Customer Storefront	tier0	Restore-tested	RP-WEB-202610200200-P	7 h	met (1 d)	n/a	not applicable (3 h)	65
Internal ERP	tier1	Restore-tested	RP-ERP-202610200400-P	5 h	breached (4 h)	n/a	not applicable (8 h)	35
Corporate File Storage	tier2	Integrity only	RP-FILES-202610200100-P	8 h	met (1 d)	n/a	not applicable (1 d)	56
Central Logging & SIEM	tier1	Restore-tested	RP-SIEM-202610200700-P	2 h	breached (1 h)	n/a	not applicable (4 h)	80
Monitoring & Alerting	tier2	Integrity only	RP-MON-202610200000-P	9 h	met (1 d)	n/a	not applicable (8 h)	76
HR Self-Service Portal	tier3	Integrity only	RP-HR-202610192300-P	10 h	unknown (undoc.)	n/a	not applicable (undoc.)	36
Analytics Export Bucket (AWS S3)	tier3	No usable point	none eligible	not bounded	breached (1 d)	n/a	not applicable (2 d)	15

Score = criticality-weighted mean of per-system scores: backup confidence 35, RPO 20, RTO in a full-recovery rehearsal 20, dependencies 15, isolation 10. Unknown values earn no credit.

5. Recovery dependency order

Estimated recovery time: Cannot be estimated: every recovery step is blocked. Potential data-loss exposure: 7 h; no usable recovery point for db-orders (loss not bounded).

#	Wave	System	Prerequisites	Recovery point	Start	Ready	Basis	RTO
1	-	Orders Database (PostgreSQL)	-	-	BLOCKED		cannot be restored: Backup encryption key disabled and pending deletion; recovery points cannot be decrypted	breached
2	-	Orders API Service	db-orders	RP-ORDAPI-202610200200-P	BLOCKED		waiting on blocked prerequisite(s): db-orders	breached

- Recovery starts when the incident is declared (T+0). Detection, triage, forensic hold and decision time are not included in estimates.
- The parallel estimate assumes unlimited restore capacity, so independent assets restore simultaneously; the serial estimate assumes one asset at a time. Reality lies between the two.
- Restore durations come from the most recent passing restore test where one was measured; otherwise the documented estimate is used. Unknown durations are not guessed.
- An asset may start restoring only after every prerequisite in the plan has finished.
- Assets outside the plan scope are assumed to remain operational.

6. Findings and remediation recommendations

[CRITICAL] Unrecoverable dependency blocks dependent systems - Orders Database (PostgreSQL)

F-DEP-003-db-orders | rule DEP-003 | basis: simulated | priority 1

Orders Database (PostgreSQL) cannot be restored (cannot be restored: Backup encryption key disabled and pending deletion; recovery points cannot be decrypted). 3 dependent system(s) cannot return to service: app-erp, app-orders-api, web-storefront. app-orders-api have intact, eligible backups of their own, but restoring them alone is insufficient while this prerequisite is unavailable.

Calculation: Transitive dependents: 3; criticality-weighted impact: 11 (tier0=4, tier1=3, tier2=2, tier3=1).

Business impact: Tier 0 - Mission critical. 3 dependent system(s): Internal ERP, Orders API Service, Customer Storefront.

Action: Restore or validate the critical dependency first (alternative recovery source, key recovery, or rebuild), then resume the dependent recovery. Verify by: Demonstrate a successful restore of the dependency in an isolated environment. Expected outcome: Dependency becomes restorable and dependents are unblocked.

Evidence: EV-RST-DBORD-202610120900

[CRITICAL] No usable recovery point - Orders Database (PostgreSQL)

F-RP-001-db-orders | rule RP-001 | basis: simulated | priority 2

No eligible recovery point for Orders Database (PostgreSQL). RP-DBORD-202610200845-P: restore blocked: Backup encryption key disabled and pending deletion; recovery points cannot be decrypted; RP-DBORD-202610200830-P: restore blocked: Backup encryption key disabled and pending deletion; recovery points cannot be decrypted; RP-DBORD-202610200815-P: restore blocked: Backup encryption key disabled and pending deletion; recovery points cannot be decrypted; RP-DBORD-202610200100-I: restore blocked: Backup encryption key disabled and pending deletion; recovery points cannot be decrypted.

Business impact: Tier 0 - Mission critical. 3 dependent system(s): Internal ERP, Orders API Service, Customer Storefront.

Action: Establish a working backup for this asset and produce at least one integrity-verified recovery point in an isolated repository. Verify by: Run a backup job, record a passing integrity check, and confirm the point appears as eligible in RecoveryProof. Expected outcome: Asset shows an eligible recovery point and RP-001 is no longer raised.

Evidence: EV-INT-DBORD-202610120200-P, EV-INT-DBORD-202610190645-P, EV-INT-DBORD-202610200100-I, EV-INT-DBORD-202610200815-P, EV-INT-DBORD-202610200830-P, EV-INT-DBORD-202610200845-P, EV-RST-DBORD-202610120900

[HIGH] Unapproved roles can delete recovery points - Primary Backup Vault

F-ACC-001-vault-primary | rule ACC-001 | basis: verified | priority 3

Unapproved role(s) can delete recovery points in Primary Backup Vault: platform-admin. Approved: backup-admin.

Business impact: Tier 0 - Mission critical.

Action: Reduce backup-administrator privileges: remove delete permissions from unapproved roles and require multi-party approval for deletion. Verify by: Perform an access review and record it as evidence. Expected outcome: Only approved roles can delete recovery points.

Evidence: EV-ACC-PRIMARY

[HIGH] Shared critical dependency lacks a restore test - Orders Database (PostgreSQL)

F-DEP-004-db-orders | rule DEP-004 | basis: derived | priority 4

3 critical system(s) depend on Orders Database (PostgreSQL), whose recovery evidence is 'No usable point'.

Business impact: Tier 0 - Mission critical. 3 dependent system(s): Internal ERP, Orders API Service, Customer Storefront.

Action: Restore or validate this critical dependency with a full restore test and document its recovery runbook. Verify by: Record a passing restore test for the dependency. Expected outcome: Dependency is restore-tested.

[HIGH] Contradictory evidence - Identity Provider & SSO

F-EV-002-idp-core-RP-IDP-202610200730-P | rule EV-002 | basis: verified | priority 5

RP-IDP-202610200730-P: reported integrity check status contradicts evidence; treated as 'failed'.

Business impact: Tier 0 - Mission critical. 6 dependent system(s): Internal ERP, Orders API Service, Corporate File Storage, HR Self-Service Portal, Monitoring & Alerting, Customer Storefront.

Action: Investigate the contradiction between the backup platform and the evidence store; re-run verification. Verify by: Re-verify the recovery point and confirm reported status and evidence agree. Expected outcome: Reported status and evidence agree.

Evidence: EV-INT-IDP-202610200730-P

[HIGH] No recent end-to-end restore test - Customer Database (PostgreSQL)

F-RP-005-db-customers | rule RP-005 | basis: verified | priority 6

No passing end-to-end restore test within 90 days for Customer Database (PostgreSQL) (last passing test 2026-06-15 13:00 UTC, 126 days before assessment).

Business impact: Tier 0 - Mission critical. 1 dependent system(s): Customer Storefront.

Action: Schedule and perform an end-to-end restore test into an isolated environment, including application-level validation. Verify by: Record the restore test result and measured duration as evidence. Expected outcome: Asset recovery evidence becomes 'Restore-tested' with a measured duration.

Evidence: EV-RST-DBCUS-202606151300

[HIGH] No verified isolated recovery copy - Internal ERP

F-ISO-001-app-erp | rule ISO-001 | basis: verified | priority 7

Tier 1 - Business critical assets require an isolated, immutable copy; isolation status is 'none'.

Business impact: Tier 1 - Business critical.

Action: Create an isolated backup copy in a separate account/tenant with compliance-mode immutability. Verify by: Confirm a recovery point exists in the isolated repository and that its immutability configuration is evidenced. Expected outcome: Isolation status becomes 'verified'.

[HIGH] Recovery point objective breached - Internal ERP

F-RP-002-app-erp | rule RP-002 | basis: derived | priority 8

Newest eligible recovery point for Internal ERP is 5 h old at the time of the incident; RPO is 4 h.

Calculation: Incident 2026-10-20 09:00 UTC - eligible point RP-ERP-202610200400-P 2026-10-20 04:00 UTC = 5 h exposure; RPO 4 h; gap 1 h.

Business impact: Tier 1 - Business critical. Potential data loss of 5 h exceeds tolerance by 1 h.

Action: Correct the backup schedule or failing jobs so that the backup interval is shorter than the RPO, and alert on missed backups. Verify by: Review backup job logs for the last 7 days and confirm a verified recovery point exists within the RPO window. Expected outcome: Data-loss exposure is at or below the RPO.

[HIGH] Recovery point objective breached - Central Logging & SIEM

F-RP-002-logging-siem | rule RP-002 | basis: derived | priority 9

Newest eligible recovery point for Central Logging & SIEM is 2 h old at the time of the incident; RPO is 1 h.

Calculation: Incident 2026-10-20 09:00 UTC - eligible point RP-SIEM-202610200700-P 2026-10-20 07:00 UTC = 2 h exposure; RPO 1 h; gap 1 h.

Business impact: Tier 1 - Business critical. Potential data loss of 2 h exceeds tolerance by 1 h.

Action: Correct the backup schedule or failing jobs so that the backup interval is shorter than the RPO, and alert on missed backups. Verify by: Review backup job logs for the last 7 days and confirm a verified recovery point exists within the RPO window. Expected outcome: Data-loss exposure is at or below the RPO.

[MEDIUM] Retention lock can be bypassed by privileged roles - Primary Backup Vault

F-ISO-003-vault-primary | rule ISO-003 | basis: verified | priority 10

Governance-mode retention on Primary Backup Vault can be bypassed by: backup-admin, platform-admin.

Business impact: Tier 0 - Mission critical.

Action: Move critical recovery points to compliance-mode retention or remove bypass permissions from day-to-day administrator roles. Verify by: Review the repository policy and confirm bypass permissions are restricted to a break-glass role. Expected outcome: No standing role can bypass retention.

Evidence: EV-IMM-PRIMARY

[MEDIUM] Retention below policy minimum - Customer Database (PostgreSQL)

F-RET-001-db-customers | rule RET-001 | basis: verified | priority 11

Retention is 14 days; policy minimum for tier0 is 35 days.

Calculation: 14 d < 35 d required

Business impact: Tier 0 - Mission critical. 1 dependent system(s): Customer Storefront.

Action: Improve recovery-point retention to meet the policy minimum. Verify by: Confirm the retention setting and the oldest available recovery point. Expected outcome: Retention meets policy minimum.

[MEDIUM] Status claimed without supporting evidence - Central Logging & SIEM

F-EV-001-logging-siem-RP-SIEM-202610200800-P | rule EV-001 | basis: unknown | priority 12

RP-SIEM-202610200800-P: backup platform reports the integrity check as passed, but no evidence record supports it.

Business impact: Tier 1 - Business critical.

Action: Investigate missing backup evidence and ensure verification results are exported to the evidence store. Verify by: Locate or regenerate the evidence record and link it to the recovery point. Expected outcome: Claim is backed by an evidence record.

[MEDIUM] Newest recovery point integrity not verified - Internal ERP

F-RP-004-app-erp | rule RP-004 | basis: unknown | priority 13

Newest recovery point RP-ERP-202610200800-P (2026-10-20 08:00 UTC) has integrity status 'not_run' and is not counted as recoverable.
Business impact: Tier 1 - Business critical.
Action: Enable automated integrity verification (checksum/test mount) on every recovery point and store the results as evidence. Verify by: Confirm an integrity-check evidence record with result 'pass' exists for the newest point. Expected outcome: Newest point is integrity-verified and selected for recovery.

[MEDIUM] Newest recovery point integrity not verified - Central Logging & SIEM

F-RP-004-logging-siem | rule RP-004 | basis: unknown | priority 14

Newest recovery point RP-SIEM-202610200800-P (2026-10-20 08:00 UTC) has integrity status 'unknown' and is not counted as recoverable.
Business impact: Tier 1 - Business critical.
Action: Enable automated integrity verification (checksum/test mount) on every recovery point and store the results as evidence. Verify by: Confirm an integrity-check evidence record with result 'pass' exists for the newest point. Expected outcome: Newest point is integrity-verified and selected for recovery.

[MEDIUM] No recent end-to-end restore test - Corporate File Storage

F-RP-005-file-share | rule RP-005 | basis: unknown | priority 15

No passing end-to-end restore test within 90 days for Corporate File Storage (no passing restore test on record).
Business impact: Tier 2 - Important. 1 dependent system(s): HR Self-Service Portal.
Action: Schedule and perform an end-to-end restore test into an isolated environment, including application-level validation. Verify by: Record the restore test result and measured duration as evidence. Expected outcome: Asset recovery evidence becomes 'Restore-tested' with a measured duration.

[MEDIUM] No recent end-to-end restore test - Monitoring & Alerting

F-RP-005-monitoring | rule RP-005 | basis: unknown | priority 16

No passing end-to-end restore test within 90 days for Monitoring & Alerting (no passing restore test on record).
Business impact: Tier 2 - Important.
Action: Schedule and perform an end-to-end restore test into an isolated environment, including application-level validation. Verify by: Record the restore test result and measured duration as evidence. Expected outcome: Asset recovery evidence becomes 'Restore-tested' with a measured duration.

[MEDIUM] No usable recovery point - Analytics Export Bucket (AWS S3)

F-RP-001-aws-s3-exports | rule RP-001 | basis: unknown | priority 17

Analytics Export Bucket (AWS S3) has no backup policy and no recovery points.
Business impact: Tier 3 - Supporting.
Action: Establish a working backup for this asset and produce at least one integrity-verified recovery point in an isolated repository. Verify by: Run a backup job, record a passing integrity check, and confirm the point appears as eligible in RecoveryProof. Expected outcome: Asset shows an eligible recovery point and RP-001 is no longer raised.

[LOW] Backup encryption absent or undocumented - Corporate File Storage

F-ENC-001-file-share | rule ENC-001 | basis: unknown | priority 18

Backup encryption is not documented for confidential data.
Business impact: Tier 2 - Important. 1 dependent system(s): HR Self-Service Portal.
Action: Enable encryption at rest for backups using a managed key with restricted access, and document the configuration. Verify by: Record an encryption-configuration evidence record. Expected outcome: Backup encryption is enabled and evidenced.

[LOW] No recent end-to-end restore test - HR Self-Service Portal

F-RP-005-hr-portal | rule RP-005 | basis: unknown | priority 19

No passing end-to-end restore test within 90 days for HR Self-Service Portal (no passing restore test on record).
Business impact: Tier 3 - Supporting.
Action: Schedule and perform an end-to-end restore test into an isolated environment, including application-level validation. Verify by: Record the restore test result and measured duration as evidence. Expected outcome: Asset recovery evidence becomes 'Restore-tested' with a measured duration.

[LOW] RTO or RPO not documented - HR Self-Service Portal

F-RTO-002-hr-portal | rule RTO-002 | basis: unknown | priority 20

RTO and RPO not documented for HR Self-Service Portal.
Business impact: Tier 3 - Supporting.
Action: Document a missing RTO and RPO with the business owner. Verify by: Confirm RTO and RPO values are recorded in the inventory. Expected outcome: RTO/RPO compliance can be evaluated.

[LOW] Restore duration unknown - HR Self-Service Portal

F-RTO-003-hr-portal | rule RTO-003 | basis: unknown | priority 21

No measured or estimated restore duration for HR Self-Service Portal.
Business impact: Tier 3 - Supporting.
Action: Perform a restore test to measure the restore duration. Verify by: Record a restore test with a measured duration. Expected outcome: Restore duration is measured.

7. Evidence references

EV-ACC-PRIMARY, EV-IMM-PRIMARY, EV-INT-DBORD-202610120200-P, EV-INT-DBORD-202610190645-P, EV-INT-DBORD-202610200100-I, EV-INT-DBORD-202610200815-P, EV-INT-DBORD-202610200830-P, EV-INT-DBORD-202610200845-P, EV-INT-IDP-202610200730-P, EV-RST-DBCUS-202606151300, EV-RST-DBORD-202610120900

8. Unverified assumptions

- Analytics Export Bucket (AWS S3): restore duration unknown.
- Corporate File Storage: restore duration 1 d 6 h is an estimate, not a measurement.
- HR Self-Service Portal: RTO/RPO undocumented; compliance not assessed.
- HR Self-Service Portal: restore duration unknown.
- Monitoring & Alerting: restore duration 1 h is an estimate, not a measurement.
- All infrastructure, backup and evidence data in this analysis is synthetic unless the data source states otherwise.
- A recovery point counts as recoverable only when an evidence record shows a passing integrity check; reported status without evidence is treated as unknown.
- Restore-tested means a passing end-to-end restore test exists within the policy window; it validates the procedure, not every individual recovery point.
- Unknown values earn no score credit and are reported as 'Unknown' rather than assumed.
- Data-loss exposure is the time between the incident and the newest eligible recovery point; transactions in flight and replication lag are not modelled.
- All timestamps are UTC.

9. Limitations and confidence

Moderate confidence: 7/13 systems have restore-tested evidence; 1 RPO and 0 RTO values could not be evaluated. Report generated 2026-10-28 07:09 UTC from analysis version 2026.10.1.

- RecoveryProof is a decision-support tool. It does not perform, guarantee or replace tested enterprise backup and recovery.
- Conclusions are only as good as the evidence supplied; missing evidence is reported as unknown, never as success.
- Time estimates exclude detection, triage, forensic preservation and business validation, and assume restore capacity as stated in the plan assumptions.
- The local audit trail is tamper-evident (hash-chained), not tamper-proof.